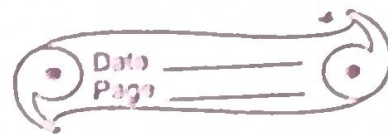


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DAA Assignment

Q.1 Find the max profit for rod cutting problem

having length = 1 2 3 4 5 6 7 8

Price = 1 5 6 9 11 12 14 18

Ans

		j → length									
		0	1	2	3	4	5	6	7	8	
j	0	0	0	0	0	0	0	0	0	0	
↓	1	0	1	2	3	4	5	6	7	8	
Part	2	0	1	5	6	10	11	15	16	20	
	3	0	1	5	6	10	11	15	16	20	
	4	0	1	5	6	10	11	15	16	20	
	5	0	1	5	6	10	11	15	16	20	
	6	0	1	5	6	10	11	15	16	20	
	7	0	1	5	6	10	11	15	16	20	
	8	0	1	5	6	10	11	15	16	20	

(2, 6)

Q.2 Solve the recurrence with master method.

Ans

$$T(n) = 4T(n/2) + n^2 \log n$$

$$a = 4 \quad b = 2$$

$$f(n) = n^2 \log n$$

$$n \log_2^4 = n^2$$

$$f(n) = n^2 \log n$$

$$c < 1$$

$$af(n/b) \leq cf(n)$$

$$4(n/2)^2 \log(n/2) \leq cn^2 \log n$$

$$n^2 \log(n/2) \leq cn^2 \log n$$

$$\log(n/2) \leq c \log n$$

$$\log n - \log 2 \leq c \log n$$

$$T(n) = O(f(n))$$

$$T(n) = O(n^2 \log n) \text{ Ans}$$

Q.3 Find min cost of multiplying the matrices
 $A_1 8 \times 10$ $A_2 10 \times 3$ $A_3 3 \times 12$ $A_4 12 \times 5$ $A_5 5 \times 50$

Ans

$$A_1 = 10 \times 5 \quad A_4 = 10 \times 5$$

$$A_2 = 5 \times 2$$

$$A_3 = 20 \times 10$$

$$A_1 ((A_2 A_3) A_4)$$

$$A_2 A_3 = 5 \times 20 \times 10 = (5 \times 10) = 1000$$

$$(A_2 A_3) A_4 = 5 \times 10 \times 5 = (5 \times 5) = 250$$

$$A_1 ((A_2 A_3) A_4) = 10 \times 5 \times 5 = (10 \times 5) = 250$$

• minimum no. of scalar multiplication required

$$= 1000 + 250 + 250 = 1500 \text{ Ans}$$

Q.4

$$(x_1 \vee x_2) \wedge (x_2 \vee \bar{x}_1) \wedge (x_1 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2)$$

x_1 True

x_2 False

Solve the 2-SAT problem

Ans

$$(x_1 \vee x_2) \wedge (x_2 \vee \bar{x}_1) \wedge (x_1 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2)$$

$x_1 = \text{True}$

$x_2 = \text{False}$

1.

$$(x_1 \vee x_2) = (\text{True} \vee \text{False}) = \text{True}$$

2.

$$(x_2 \vee \bar{x}_1) = (\text{False} \vee \text{False}) = \text{False}$$

3.

$$(x_1 \vee \bar{x}_2) = (\text{True} \vee \text{True}) = \text{True}$$

4.

$$(\bar{x}_1 \vee x_2) = (\text{False} \vee \text{False}) = \text{False}$$

$$\text{True} \wedge \text{False} \wedge \text{True} \wedge \text{False} = \text{False}$$

It does not satisfy the 2-SAT Problem.

Q.5

Find the max profit for given knapsack problem. $|A| = 65$.

Ans

w	6	11	12	10	25	30	25	15
p	11	12	23	15	20	19	26	18

$$|A| = 65$$

item - 3 :- $w=12$, $P=23$, $TW=12$, $TP=23$

item - 4 :- Weight = 10, Profit = 15, Total $w=28$, $TP=48$

item - 1 :- $w=6$, $P=11$, $TW=18$, $TP=34$

item - 8 :- $w=15$, $P=18$, $TW=43$, $TP=67$

item - 2 :- $w=11$, $P=12$, $TW=54$, $TP=79$

item - 5 :- $w=25$, $P=20$, $TW=79$, $TP=99$

(skip) item - 7 :- $w=55$, $P=26$, $TW=89$, $TP=105$

item - 6 :- $w=30$, $P=19$, $TW=84$, $TP=105$

$TW=84$, $P=79$ Ans

Q17 list and briefly define categories of security mechanisms.